

EC-350 AI and Decision Support Systems

Week 1 Introduction

Dr. Arslan Shaukat



Acknowledgement: Lecture slides material from
Stuart Russell

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Introduction

- Education
 - *BE and MS Computer Engineering*
 - College of EME, NUST
 - *PhD Computer Science (Machine Learning)*
 - The University of Manchester, U. K.
- Experience
 - *More than 12 years*



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Introduction

- Published 60+ research articles in various journals and conferences
- Successfully supervised various UG and PG students
- Head PG Program of Dept. of C&SE
- Received Best Teacher Award of CEME, NUST, 2018
- Received Best Conference Paper Award at IEEE ITEC-AP Conference, South Korea, 2019



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Research Areas of Interest

- AI and Machine Learning
 - *Feature Extraction and Selection*
 - *Segmentation, Classification, Prediction*
 - *CNN and Deep Learning*
- Digital Image and Speech Processing
- Bio Medical Imaging
- Facial and Speech Emotion Recognition



From: <https://www.aab.ca/web/newsroom/news-detail/11-11-like-you-more-if-i-can-guess-your-facial-expression-1345668003610.html?articleid=1345752005033>

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Research Projects

- Automated Approach for Patient Experience Management System
- Intelligent Human Speech Impersonation Detection for Remote Authentication
- Assessment of Academic Maturity among Students from Purpose Statements
- Diagnosis and Prognosis of COVID-19 Using Volumetric Three-Dimensional Computed Tomography Images
- Citizen Complaints Classification using NLP and Deep Learning
- Blind Image Deblurring using Generative Adversarial Network
- Automatic Classification of White Blood Cell Images using CNN
- Emotion Recognition from Facial Images using Hybrid Deep Learning
- Automatic Prostate Cancer Grading Using Deep Architectures
- Detection and Localization of Pneumothorax from Imbalanced X-ray Images Dataset Based on Deep Learning Methods

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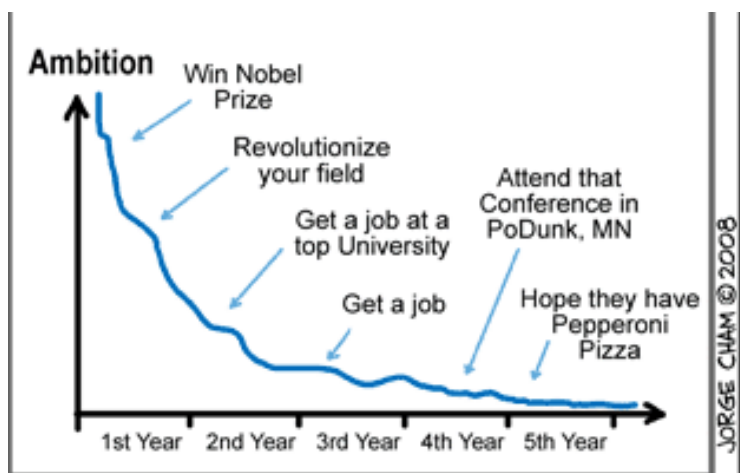
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Introduction – (Yours)

– *Aims and goals for life?*



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Introduction – (Yours)

- Name
- Previous institution/City
- Area/Subject of interest
- Where you see yourself in future



From: <https://www.vectorstock.com/royalty-free-vector/business-dreams-future-planning-vector-1739559>

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Course Information

- Course Title: AI and Decision Support Systems
- Course Code: EC-350
- Credit Hours: 3-1
- Semester: Fall 2022
- Email: arslan.shaukat@ce.ceme.edu.pk
- Class: Tue & Wed

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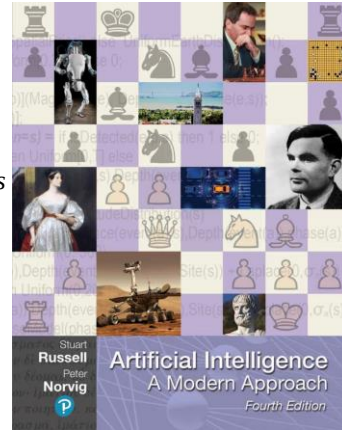
Books

- Textbook:

- *S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, 2020*

- Reference:

- *Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th ed. G. Luger, Addison Wesley, 2009*
- *Artificial Intelligence: A Systems Approach. M. Tim Jones, Infinity Science Press, 2008*
- *Pattern Classification (2nd edition 2006), by Richard O. Duda, Peter E. Hart and David G. Stork, Wiley Inter-science.*
- *Pattern Recognition and Machine Learning by Christopher Bishop, Springer 2006.*



Course Contents

- Introduction to Artificial Intelligence
- Agents, PEAS model, Rationality, Nature & properties of environment
- Structures of agents
- Problem solving by searching, Uninformed search strategies, Breadth first search (BFS)
- Depth first search (DFS) , Depth limited search, Iterative deepening DFS
- Informed search strategies, Greedy best first search, A* search
- Genetic algorithms
- Games, Minimax algorithm, Alpha beta pruning

Course Contents

- Introduction to Machine Learning and basic types of classifiers, Performance parameters for evaluation
- K-Nearest Neighbor (KNN), ROC Analysis
- Bayesian Decision Theory
- Naïve Bayes Classifier
- Neural networks and single layer Perceptron
- Multi layer Neural Networks
- Introduction to decision trees

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Marks Distribution

- **Lecture 75%**
 - *Mid Term Exam* 30%
 - *Final Exam* 45%
 - *Quizzes* 15%
 - *Assignments* 10%

- **Lab 25%**
 - *Lab* 70%
 - *Project* 30%

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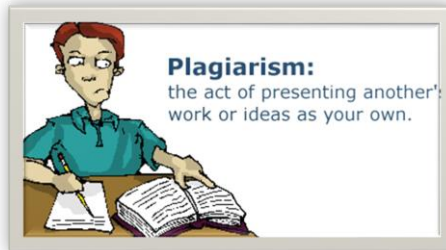
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Policies

- No extensions in assignment deadlines.
- Quizzes will be unannounced.
- Never cheat.
 - “Better fail NOW or else will fail somewhere LATER in life”
- Plagiarism will also have strict penalties.



Adapted from What is Plagiarism PowerPoint
<http://mciu.org/~spjvweb/plagiarism.ppt>

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Course Learning Outcomes

Course Learning Outcomes (CLOs)	Level of Learning	PLO
1. Solving problems using various uninformed and informed search strategies.	C3	2
2. Apply local search algorithms like Genetic Algorithm (GA) on optimization problems and perform Minimax search on games.	C3	2
3. Design machine learning systems, demonstrating understanding of machine learning concepts including feature extraction and classification.	C6	3
4. Implement projects in the lab work that use Python and MATLAB for execution of the theoretical knowledge gained during class lectures, requiring some independent reading, programming and learning.	P2	12

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Artificial Intelligence (AI)

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What is Intelligence?

- A machine is intelligent if it has:
 - *Ability to learn and adapt from environment*
 - *Ability to acquire knowledge*
 - *Ability to memorize and proceed*
 - *Ability to plan and schedule*
 - *Ability to recognize voice, patterns, faces*
 - *Ability to understand and perceive*
 - *Ability to solve complex problems*



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What is Artificial Intelligence

- If a machine could do all or most of the mentioned tasks, we can call that machine an intelligent machine
- The ‘intelligence’ in the machine is NOT natural but artificial
- Name was coined in 1956.

What is Artificial Intelligence

- Concerned with *building* intelligence in artificial – man made devices
- Making machines to behave like humans – if we consider humans to be intelligent
- Making machines to behave in most rational manner
 - *Thinking/reasoning intelligently*
 - *Acting/behaving intelligently*

Approaches to AI

- Think
 - *Systems that “think” like humans*
 - *Systems that “think” rationally*
- Act
 - *Systems that “act” like humans*
 - *Systems that “act” rationally*

To Think

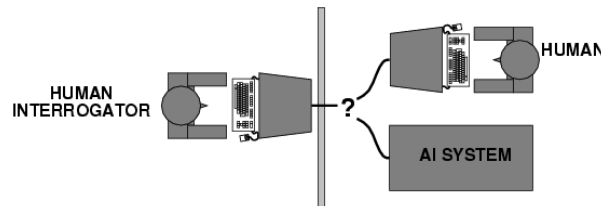
- Make machines to think intelligently [like human beings] as if they have real conscious minds
- Deals with “How brain works”
- Actually tries to recreate the functions of the inside of the brain
- Advocates Machine can replace a human being
- Many researchers believe it is NOT possible to attain

To Act

- Machines can be made to act as if they are intelligent
- Treats brain as a black box which creates output after it receives input
- Is NOT concerned with inner functionality of brain
- Most of the research is done in this field

Acting Humanly – Turing Test

- Alan Turing (1950) “Computing Machinery and intelligence” at The University of Manchester
- Can Machine behave intelligently?
- Operational test for intelligent behavior



Turing Test

- Suggested major components of AI
 - *natural language processing*
 - to enable it to communicate successfully in English
 - *knowledge representation*
 - to store what it knows or hears
 - *automated reasoning*
 - to use the stored information to answer questions and to draw new conclusions
 - *machine learning*
 - to adapt to new circumstances and to detect and extrapolate patterns.
 - *computer vision*
 - to perceive objects
 - *robotics*
 - to manipulate objects and move about.

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Thinking Humanly – Cognitive Science

- A system is intelligent if it *thinks* like a human.
- If a system uses the same reasoning processes as a human, then it is intelligent
- Requires scientific theories of internal working of brain
- Three ways:
 - *Try to catch our own thoughts as they go by*
 - *Through psychological experiments*
 - *Observing the brain in action*
- Cognitive Science brings together computer models from AI and experimental techniques from psychology
- To construct theories of the workings of human mind

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Thinking Rationally – Laws of Thought

- A system is intelligent if it thinks *rationally*.
- An intelligent system is one that follows sound reasoning processes that always lead to correct outcomes.
- Logicians in the 19th century developed a precise notation for statements about all kinds of objects in the world and the relations among them.
- This leads to the study of logic and formal reasoning
- However, logic has its problems.
 - *Formalizing common-sense knowledge*
 - *Computational issues*

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Acting Rationally

- A system is intelligent if it *acts rationally*.
- A *rational agent* is one that acts to achieve the best possible (expected) outcome, given its knowledge and ability.
- We can build agents that do certain tasks intelligently without having human-level intelligence.
- All the skills needed for Turing Test also allow an agent to act rationally.

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Foundations of AI

- Many disciplines provided the foundation for Artificial Intelligence. Few to mention are as follows:
- Philosophy (428 BC – Present)
 - *Made AI conceivable by considering the ideas that the mind is in some ways like a machine*
- Mathematics (800 – Present)
 - *Set the groundwork for understanding computation and reasoning about algorithms.*
- Economics (1776 – Present)
 - *Formalized the problem of making decisions that maximize the expected outcome to the decision maker.*

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Foundations of AI

- Neuroscience (1861 – Present)
 - *Discovered some facts about how the brain works and the ways in which it is similar to and different from computers.*
- Psychology (1879 – Present)
 - *Studied about humans and animals thinking and actions*
- Computer Engineering (1940 – Present)
 - *Provided the ever-more-powerful machines that make AI applications possible.*
- Control theory & Cybernetics (1948 – Present)
 - *Deals with designing devices that act optimally on the basis of feedback from the environment.*
- Linguistics (1957 – Present)
 - *Showed how does language relate to thought.*

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Few Applications using AI

- Game playing
- Robotics and robot vehicles
- Tutoring systems
- Medicine
- Biometrics
- Speech Recognition
- Autonomous planning and Scheduling
- Spam fighting



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Few Applications using AI

- Deep Blue
 - <http://www-03.ibm.com/ibm/history/ibm100/us/en/icons/deepblue/>
- Aaron the Robot as an Artist
 - <http://aaronshome.com/aaron/index.html>
- Speech recognition
 - <http://www.nuance.com/>
- Honda Humanoid Robot
 - <https://global.honda/innovation/robotics/ASIMO.html>
- Mars rover curiosity
 - <http://mars.jpl.nasa.gov/msl/mission/overview/>

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Few Applications using AI

- Microsoft
 - <https://www.microsoft.com/en-us/research/research-area/artificial-intelligence/>
- Google
 - <https://ai.google/>
- Association for the Advancement of AI
 - <http://www.aaai.org>
- AI Book
 - <http://aima.cs.berkeley.edu/>

Intelligent Agents

Intelligent Agent

- An *agent* is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.
- Agent Examples
 - *A human agent has eyes, ears, and other organs for sensors and hands, legs, mouth and other body parts for actuators*
 - *A robotic agent substitutes cameras and infrared range finders for the sensors and various motors for the actuators.*
 - *A software agent receives keystrokes, network packets, file contents as sensory input and acts upon the environment by displaying on screen, sending network packets and writing files.*

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Percept & Percept Sequence

- Percept refers to agent's perceptual input at any given instance
- Percept sequence is the complete history of everything agent has ever perceived
 - *An agent's choice of action at any given instance can depend on the entire percept sequence observed to-date*

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Agent Function

- Mathematically, an agent's behavior is described by the agent function that maps any given percept sequence to an action
 - For most agent this would become a very large table. May be of infinite size
 - Have to put a bound on the length of the percept sequence we want to consider
 - Internally, agent function for an artificial agent shall be an agent program
 - Agent Function is a mathematical description
 - Agent Program is a concrete implementation

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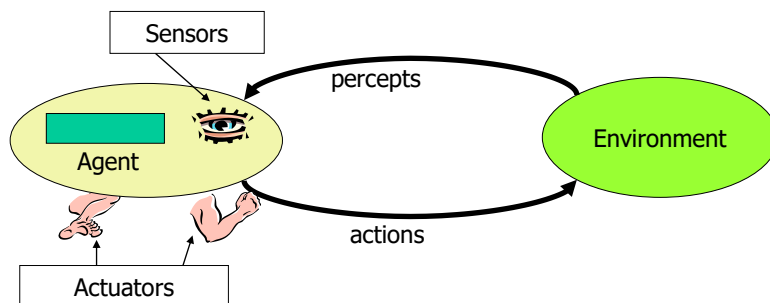
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An Intelligent Agent



The **agent function** maps from percept histories to actions:

$$f: \mathcal{P}^* \rightarrow \mathcal{A}$$

$\mathcal{P}^*$ = Percept History

$\mathcal{A}$ = Actions

The **agent program** runs on the physical **architecture** to produce f

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Vacuum-Cleaner Agent



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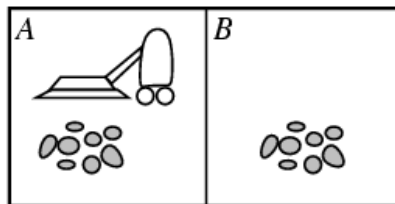
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Vacuum-Cleaner Agent



- Locations: square A, square B
- Percepts: location and contents, e.g., [A, Dirty]
- Actions: Left, Right, Suck, NoOp



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Vacuum-Cleaner Agent

Percept Sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck

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Good Behavior - Rationality

- *Rational Agent* is the one which does the “right thing” based on what it perceives and the actions it performs.
- Every entry in the table for the agent function is filled out correctly
- The right action is the one that will cause the agent to be most successful
- How can we say an agent is successful?
- We need some criterion to measure its success

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Performance Measure

- An objective criterion for success of an agent's behavior
- e.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc
- No single universal criterion for every agent
- As a general rule, it is better to design a performance measure according to what one actually wants in the environment rather than according to how one thinks the agent should behave
- The selection of performance measure is not always easy

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Rationality

- What is rational at given time depends upon four things
 - *The performance measure that defines criterion for success*
 - *The agent's prior knowledge of the environment*
 - *The actions that agent can perform*
 - *The agent's percept sequence to-date*

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Rational Agent

- For each possible *percept sequence*, a rational agent should select an *action* that is expected to maximize its *performance measure*, given the evidence provided by the percept sequence and whatever *built-in knowledge* the agent has.
- Vacuum-cleaner agent?

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Vacuum-Cleaner Agent

- The performance measure awards one point for each clean square at each time step, over a "lifetime" of 1000 time steps.
- The "geography" of the environment is known a priori but the dirt distribution and the initial location of the agent are not.
- The only available actions are Left, Right, and Suck.
- The agent correctly perceives its location and whether that location contains dirt.

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Omniscience

- An omniscience agent knows the actual outcomes of its actions and can act accordingly
- Omniscience is impossible in reality
- Road crossing example

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Omniscience, Perfection

- Rationality is NOT the same as Perfection
- Rationality maximizes expected performance
- Perfection maximizes actual performance

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Exploration, Learning

- Doing actions in order to modify future percepts, sometimes called *information gathering*, is an important part of rationality.
- It performs such actions to increase its perception
- This is also called Exploration
 - *as taken by vacuum cleaner agent*
- A rational agent not only gather information but also learn as much as possible from what it perceives

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Agent Autonomy

- The capacity to compensate for partial or incorrect prior knowledge by learning
- An agent is called autonomous if its behavior is determined by its own experience (with ability to learn and adopt)
- A truly autonomous agent should be able to operate successfully in a wide variety of environments

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